ICAO Doc 9303 Part 10: Logical Data Structure (LDS) and Contactless IC Chip Security

Type: technical_guide | **Date:** 2021-11-20

1. Logical Data Structure (LDS1) Data Groups

The contactless integrated circuit embedded in e-passports and electronic ID cards contains standardized Data Groups (DG). DG1 stores the MRZ info. DG2 contains the primary facial image template (JPEG or JPEG2000 format, minimum 480x640 pixels). DG3 holds fingerprint templates (restricted access via EAC). DG14 contains Chip Authentication public keys. DG15 contains Active Authentication public keys.

2. Access Control Mechanisms: BAC and PACE

Basic Access Control (BAC) derives encryption keys from the MRZ line (document number, date of birth, expiry date). Password Authenticated Connection Establishment (PACE / SAC) utilizes Diffie-Hellman key agreement based on the Card Access Number (CAN) or MRZ, providing robust resistance against eavesdropping.